STUDENT NUMBER:

APSC 171 - Term Test 1
Tuesday October 1, 2024

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First Name: ADA

Student #: 18151852

- This is a 2 hour test.
- No textbooks, notes, or aids other than your calculator (Casio 991 series) are allowed.
- For full marks, you must show all your work and explain how you arrived at your answers, unless explicitly told to do otherwise.
- Wherever appropriate, **include units in your answers**.
- When drawing graphs, **add labels and scales on all axes**.
- Correct final answers with incorrect support may be given no credit.
- While waiting to start, fill in your name and student number above.
- If you need more room, there are blank pages at the end of the test. **If you use these pages, you must provide clear directions on the original page**, e.g. "Continued on page 10".
- Only answers written in this booklet will be graded; all scrap paper from the test will be discarded.

Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	Total
6	7	7	6	6	8	9	6	55

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1. Differentiate the following functions with respect to their input variable.

   - You may assume that the input variables will always be within the domain of their respective functions.)
   - You do **not** need to simplify after you have taken the derivative.

(a) $f(x) = \cos\left(e^{3x}\right)$ [/2]

$f'(x) = -\sin(e^{3x}) \cdot e^{3x} \cdot 3$

or tidying, $= -3e^{3x} \sin(e^{3x})$

(b) $\vec{r}(t) = \langle \sin(t)\ln(2t),\ 4^t \rangle$ [/2]

$\vec{r}\,'(t) = \left\langle \cos(t)\ln(2t) + \sin(t)\frac{1}{2t}\cdot 2,\ 4^t \ln(4) \right\rangle$

or tidying, $= \left\langle \cos(t)\ln(2t) + \frac{1}{t}\sin(t),\ 4^t \ln(4) \right\rangle$

(c) $g(u) = \dfrac{3u^2 - 7\sqrt{u}}{u^{3/2}}$ [/2]

tidy 1st $\rightarrow = 3u^{1/2} - \frac{7}{u}$

or tidying,

so $g'(u) = \frac{3}{2}u^{-1/2} + \frac{7}{u^2} = \frac{3}{2\sqrt{u}} + \frac{7}{u^2}$

or using quotient rule:

$$= \frac{(6u - \frac{7}{2}u^{-1/2})(u^{3/2}) - (3u^2 - 7\sqrt{u})(\frac{3}{2}u^{1/2})}{u^3}$$

tidying $= \frac{6}{\sqrt{u}} - \frac{7}{2u^2} - \frac{9}{2\sqrt{u}} + \frac{21}{2u^2} = \frac{3}{2\sqrt{u}} + \frac{7}{u^2}$

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2. Consider the vector-valued function $\vec{u}(t) = \langle 3e^{-t},\ 4e^{t} \rangle$.

(a) Eliminate the parameter to find a Cartesian equation for the graph of $\vec{u}(t)$. Put your final answer in the form $y = f(x)$. [/3]

① $x = 3e^{-t}$

② $y = 4e^{t}$

longer ↓

$y/4 = e^{t}$ → $\ln(y/4) = t$ ③

③ → ① $x = 3e^{(-\ln(y/4))} = 3\,\frac{1}{e^{\ln(y/4)}}$

$e^{\ln(a)} = a$

quick → $x = 3(e^{t})^{-1}$ ① → $x = 3(y/4)^{-1}$

$e^{t} = y/4$ ②

$x = 3 \cdot \frac{4}{y} = \frac{12}{y}$

$\boxed{y = \frac{12}{x}}$

$x = \frac{3}{y/4}$ → $\frac{y}{4} = \frac{3}{x}$ → $\boxed{y = \frac{12}{x}}$

(b) Find the (x, y) coordinates of the vector-valued function at $t = 0$ and $t = 1$.
Do these points satisfy the Cartesian formula you reported in part (a), yes or no? State your conclusion clearly for the grader. [/1]

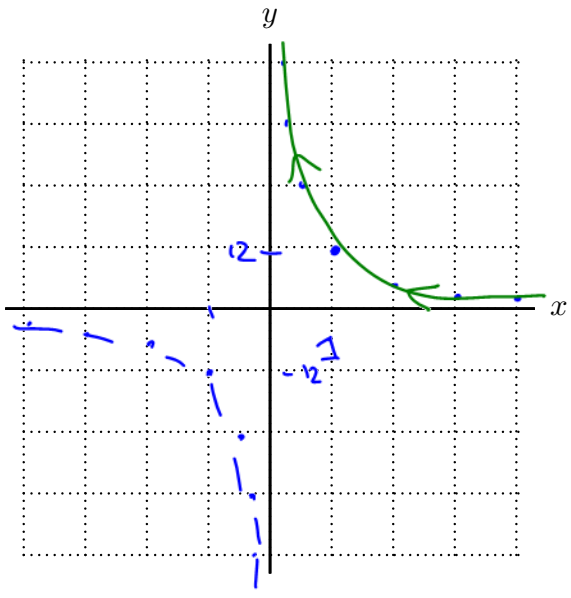
$\vec{u}(0) =$ $\langle 3, 4 \rangle$ LHS: $y = 4$; RHS $\frac{12}{3} = 4$

$\vec{u}(1) =$ $\langle 3e^{-1},\ 4e \rangle$ LHS: $y = 4e$ RHS: $\frac{12}{x} = \frac{12}{3e^{-1}} = 4e$

Points fit (a) answer (yes or no): yes!

(c) On the axes to the right, sketch the graph of the Cartesian equation you found in part (a). [/3]
Once you have drawn the graph,

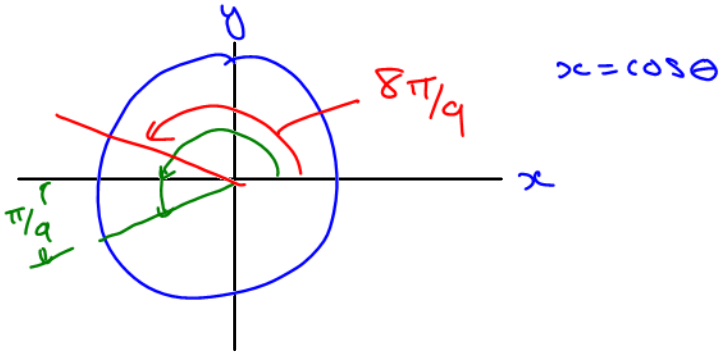
- Clearly indicate what part of this graph the function $\vec{u}$ actually passes over, and
- draw arrows to indicate the direction of motion;
- include scales on your axes.

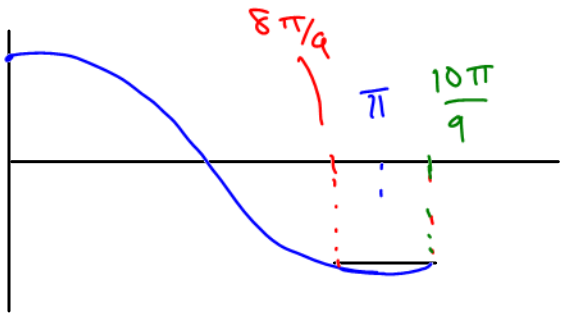


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3. The questions on this page are all independent of each other.

(a) Find the exact value of $\arccos\left(\cos\left(\frac{10}{9}\pi\right)\right)$. $= \frac{8\pi}{9}$ [/2]

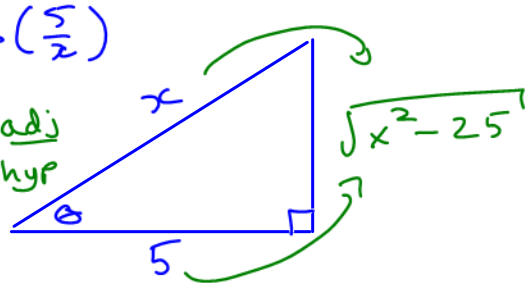




(b) Give an alternate form of the function
$f(x) = \sin\left(\arccos\left(\frac{5}{x}\right)\right)$
which does not involve trigonometric functions. (~~You may assume $-5 \le x < 0$ or $0 < x \le 5$.~~) [/2]

Let $\theta = \arccos\left(\frac{5}{x}\right)$

or $\cos\theta = \frac{5}{x} = \frac{\text{adj}}{\text{hyp}}$



$\sin\left(\arccos\left(\frac{5}{x}\right)\right)$

$= \sin(\theta) = \frac{\text{opp}}{\text{hyp}}$

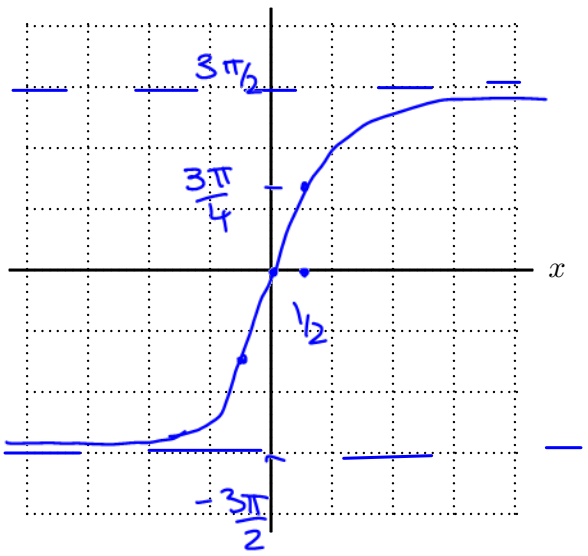
$= \boxed{\frac{\sqrt{x^2-25}}{x}}$

(c) Sketch the graph of $y = 3\arctan(2x)$ on the axes below. Indicate the scale on each axis, and also clearly indicate the locations of any horizontal and/or vertical asymptotes. [/3]

Usually, $\arctan(1) = \pi/4$, and horizontal asymptotes @ $y = \pi/2, -\pi/2$

But $3\arctan(2x)$ is

- stretched vertically by 3, and
- compressed horizontally by 2.



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4. Consider the function $y = x^3$. $= f(x)$

(a) Find the equation(s) of the tangent line(s) to $y = x^3$ which also pass through the point $(0, 1)$. [/5]

at $x=a$,

$f(x) = x^3$ | $f(a) = a^3$,

so $f'(x) = 3x^2$ | $f'(a) = 3a^2$

$\Rightarrow$ tangent line to $y = x^3$ at $x=a$ is

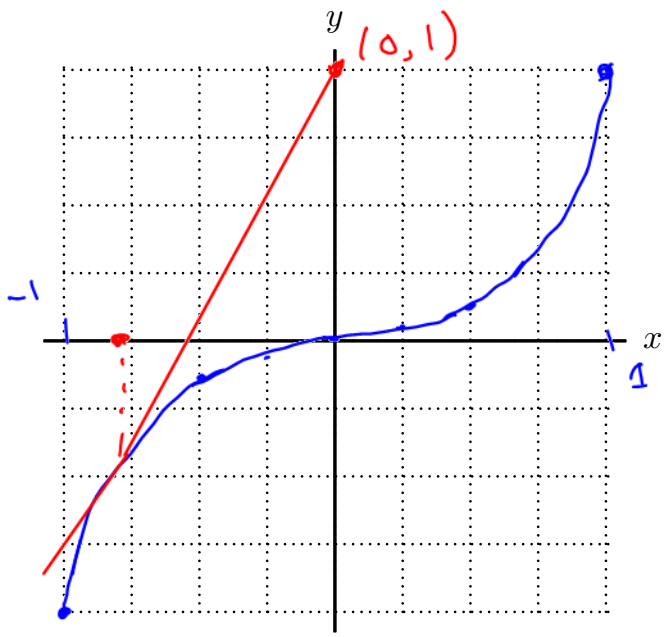
$y = \underbrace{3a^2}_{f'(a)}\underbrace{(x-a)}_{(x-a)} + \underbrace{a^3}_{f(a)}$

It passes through $(0,1) \Rightarrow$ $1 = 3a^2(0-a) + a^3$

Solving for a: $1 = -3a^3 + a^3 = -2a^3$

$a^3 = -\frac{1}{2}$ $\quad a = \frac{-1}{\sqrt[3]{2}} \doteq -0.793$

(b) Sketch the graph of $y = x^3$, and then add the tangent line(s) you found in part (a). Clearly indicate the scale you used for each axis. [/1]



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5. Two particles follow their respective trajectories given by the vector-valued functions below:

- $\vec{r}_1(t) = \langle 2t-1,\ t^2+1,\ 2t^2-5\rangle$, and
- $\vec{r}_2(t) = \langle t^2-3,\ 4+t,\ t-1\rangle$.

Determine if the particles' trajectories intersect.

- If the trajectories intersect, specify the $\langle x, y, z\rangle$ **location(s)** of the intersection in the blank spaces at the bottom of this page.
- If they do not intersect, write "No intersection point".

[/6]

→ using 's' for the $\vec{r}_2$ time:

$\vec{r}_2(s) = \langle s^2-3,\ 4+s,\ s-1\rangle$

Intersection ⇒ equal x, y, z's, but possibly at different times. We set $\vec{r}_1(t) = \vec{r}_2(s)$.

x: $2t-1 = s^2-3$ ①

y: $t^2+1 = 4+s$ ②

z: $2t^2-5 = s-1$ ③ → $s = 2t^2-4$ ④

④ → ② $t^2+1 = 4+(2t^2-4)$

$t^2+1 = 2t^2$

$1 = t^2$ → $= +1$ or -1

$s = 2t^2-4$

$s = 2-4 = -2$ $\qquad$ $s = 2-4 = -2$

Try $s=-2,\ t=1$ in ①

① LHS $= 2t-1 = 2-1 = 1$ } equal ✓

① RHS $= s^2-3 = 4-3 = 1$

$s=-2,\ t=1$ is a solution

At $t=1$, $\vec{r}_1(1) = \langle 1, 2, -3\rangle$

and

$s=-2$ $\vec{r}_2(-2) = \langle 1, 2, -3\rangle$

Try $s=-2,\ t=-1$ in ①

① LHS $= 2t-1 = -2-1 = -3$ } not equal

① RHS $= s^2-3 = 4-1 = 1$

$s=-2, t=1$ is not a solution.

Intersection location(s) (if any): $\langle 1, 2, -3\rangle$

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6. Consider a particle whose position over time is defined by $\vec{r}(t) = \left\langle \sqrt{t+1},\ \ 2 - \frac{t}{3} \right\rangle$.

We will assume lengths in m, time in s

(a) Find the particle's speed at time $t = 3$. [/2]

speed = ‖vel‖

$\vec{v}(t) = \frac{d}{dt}\vec{r}(t) = \left\langle \frac{1}{2}\frac{1}{\sqrt{t+1}}, \frac{-1}{3} \right\rangle$

At $t=3$, $\vec{v}(3) = \left\langle \frac{1}{2}\frac{1}{\sqrt{4}}, \frac{-1}{3} \right\rangle = \left\langle \frac{1}{4}, \frac{-1}{3} \right\rangle$

speed $= \sqrt{\left(\frac{1}{4}\right)^2 + \left(\frac{-1}{3}\right)^2} \simeq 0.4167$ m/s $=$ or $\frac{5}{12}$ m/s

(b) Find the particle's acceleration at time $t = 3$. [/2]

$\vec{a}(t) = \frac{d}{dt}\vec{v}(t) = \left\langle -\frac{1}{4}\frac{1}{(t+1)^{3/2}}, 0 \right\rangle$

At $t=3$, $\vec{a}(3) = \left\langle -\frac{1}{4}\frac{1}{4^{3/2}}, 0 \right\rangle = \left\langle \frac{-1}{32}, 0 \right\rangle$

(c) Find a unit vector in the direction of travel at time $t = 3$. [/2]

$= \vec{v}(3)$ (8)

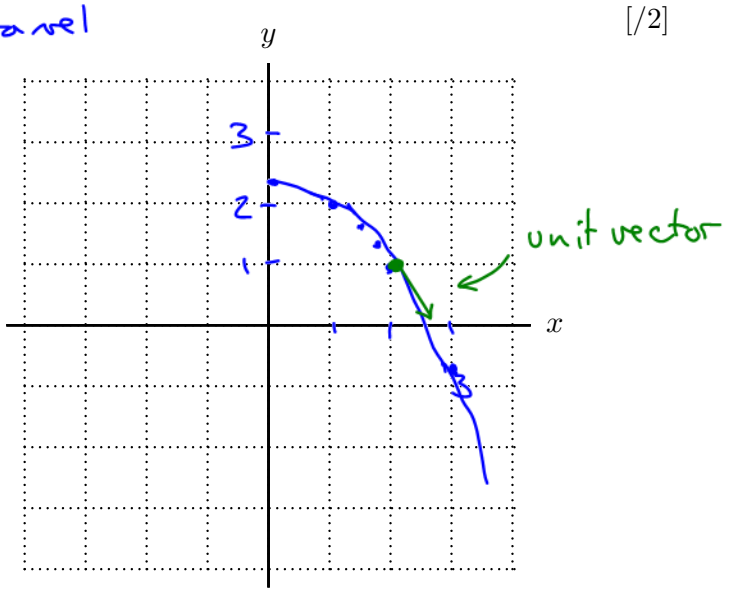
Direction is $\left\langle \frac{1}{4}, \frac{-1}{3} \right\rangle$

or $= \left\langle \frac{3}{5}, \frac{-4}{5} \right\rangle$

Length = speed $= \frac{5}{12}$ $\Rightarrow$ unit vector in dir'n of travel $= \frac{1}{5/12}\left\langle \frac{1}{4}, \frac{-1}{3} \right\rangle$

(d) On the axes to the right, sketch: [/2]

- the trajectory of the particle, including arrows to indicate the direction of motion;
- the unit vector found in part (c); and
- indicate the scale on both axes.



t	x ($\sqrt{t+1}$)	y ($2 - t/3$)
-1	0	$7/3$
0	1	2
1	$\sqrt{2}$	$5/3$
2	$\sqrt{3}$	$4/3$
3	2	1
8	3	$-2/3$

$t=3$

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7. The questions on this page are all independent of each other.

(a) Find the value(s) of k for which the vectors $\vec{u} = \langle 3, k^2 - k - 1\rangle$ and $\vec{v} = \langle 6k - 12, 4\rangle$ are **parallel**. [/3]

Parallel $\Rightarrow$ $\vec{u} = c\vec{v}$

x: $3 = c(6k-12)$ ①

y: $k^2 - k - 1 = c(4) \rightarrow c = \frac{1}{4}(k^2 - k - 1)$ ②

② → ① $3 = \frac{1}{4}(k^2-k-1)(6k-12)$

$12 = (k^2-k-1)(6k-12) = 6k^3 - 6k^2 - 6k - 12k^2 + 12k + 12$

$0 = 6k^3 - 18k^2 + 6k$

Factoring: $6k(k^2 - 3k + 1) = 0$

$k = 0$ or

$k = \frac{3 \pm \sqrt{9-4}}{2} = \frac{3 \pm \sqrt{5}}{2}$

(b) Find the value(s) of c for which the vectors $\vec{a} = \langle 2, c^2, -4\rangle$ and $\vec{b} = \langle -1, 2, 5\rangle$ are **perpendicular**. [/2]

Perpendicular $\Rightarrow$ $\vec{a} \cdot \vec{b} = 0$

$\vec{a} \cdot \vec{b} = -2 + 2c^2 - 20 \overset{\text{Set}}{=} 0$

$c^2 - 11 = 0$

$c^2 = 1$

$c = \pm\sqrt{11}$

(c) Find the value(s) of x which will make the vector $\langle 2, 3, x\rangle$ have a magnitude of exactly 10. [/2]

$\| \langle 2, 3, x \rangle \| = \sqrt{2^2 + 3^2 + x^2}$

$= \sqrt{13 + x^2} \overset{\text{Set}}{=} 10$

$13 + x^2 = 100$

$x^2 = 87$

$x = \pm\sqrt{87}$

(d) Find the angle, in radians, between the vectors $\langle -2, 3, 1\rangle$ and $\langle 5, 2, 2\rangle$. [/2]

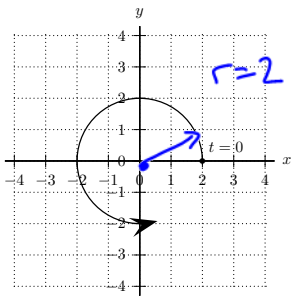
$\cos\theta = \frac{\langle -2, 3, 1\rangle \cdot \langle 5, 2, 2\rangle}{\| \langle -2, 3, 1\rangle \| \, \| \langle 5, 2, 2\rangle \|} = \frac{-10 + 6 + 2}{\sqrt{14}\sqrt{33}} \approx -0.0930$

$\Rightarrow$ $\theta \approx 1.664$ rad

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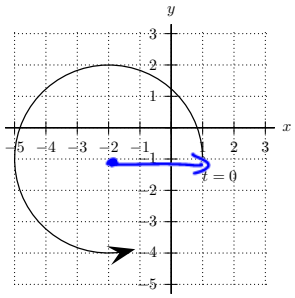
8. For each of the circles shown below, write down a parametric function that would trace out that trajectory, and match any additional properties given. No explanation of your answer is necessary on this problem.

   Assume time is measured in seconds and all distances are in meters. Multiple answers are possible; any correct answer will be accepted. [/6]



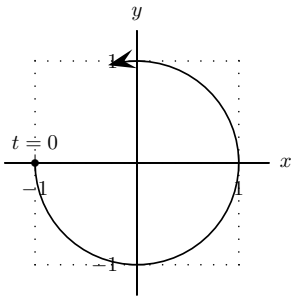
r = 2

(a) $\vec{r}(t) = \langle$ $2\cos(t)$, $2\sin(t)$ $\rangle$

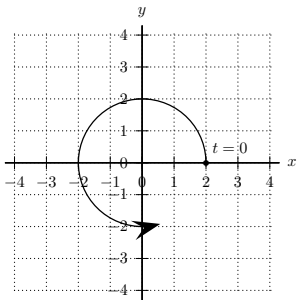


center @ (−2, −1)
r = 3

(b) $\vec{r}(t) = \langle$ $3\cos(t) - 2$, $3\sin(t) - 1$ $\rangle$



(c) $\vec{r}(t) = \langle$ $-\cos(t)$, $-\sin(t)$ $\rangle$



r = 2

(d) $\vec{r}(t) = \langle$ $2\cos\left(\sqrt{\frac{5}{2}}\,t\right)$, $2\sin\left(\sqrt{\frac{5}{2}}\,t\right)$ $\rangle$

And for this last example, (d), the uniform circular motion has a **constant centripetal acceleration of 5 m/s^2 toward the origin.**

$v = r\omega$,

$a = r\omega^2$

$5 = 2\omega^2$ $\quad \omega = \sqrt{\frac{5}{2}}$

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Space for additional work. **Indicate clearly which question you are continuing if you use this space.**

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